Zero Parameters: Deriving the Vacuum Chirality Phase from the Koide Invariant

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https://github.com/bampita-bico/CDFD-Part-I-Release

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Abstract

Paper IV derived the master self-consistency condition for the CDFT vacuum density and identified the vacuum chirality phase as the single remaining unknown. The present paper derives that phase from the Z-three algebraic structure of the trefoil vortex itself, closing the lepton-sector loop inside the model.

The argument uses a self-consistency condition. The Brannen Z-three parametrization of the lepton masses has the universal Koide invariant proved in Paper II. The chirality equilibrium condition from Paper III fixes the charged-lepton orientation once the background chirality phase is known. Self-consistency identifies the chirality-breaking angle with the Z-three invariant, setting the lepton phase to two ninths of a radian and fixing the background phase without an additional fit.

With the electron mass used only as the mass unit, the construction predicts the muon and tau masses at the reported precision and keeps the Koide relation exact inside the parametrization. The CDFT lepton-sector construction is therefore internally closed at this step, with no fitted parameter beyond the external choice of mass unit.

Across the series, the public CDFD Runtime expresses this equilibrium vacuum limit as a reusable flow-constraint state grammar. The calculations below use that equilibrium limit only; adaptive departures are left for later dynamical work rather than fitted in this manuscript. The paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, vacuum chirality phase, zero-parameter model

Project Context

This manuscript constitutes Part I (Paper 05) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 04 of this series [8].

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Symbol Conventions

CDFD physical-vacuum symbol convention.

Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

The Constraint-Driven Field Theory (CDFT) series has, paper by paper, derived the structure of the lepton sector from the geometry of the vacuum medium:

1. **Paper I** [5]: the electromagnetic fine-structure constant $\alpha = 1/137.036$ from vortex ring equilibrium.
2. **Paper II** [6]: Koide formula $Q = 2/3$ as a theorem of $\mathbb{Z}_3$ symmetry; leptons as trefoil $T(2, 3)$ modes.

3. **Paper III** [7]: All four open problems closed; ρ_0 identified as the sole remaining unknown. Chirality equilibrium: $3\theta + \phi_c = \pi$.

4. **Paper IV** [8]: Master self-consistency equation for ρ_0; Compton–ring coincidence; $M_5 =$ 460.4 MeV prediction.

In this inherited notation, α denotes the electromagnetic fine-structure constant, $\chi = R/a = 1/\alpha$ is the vortex aspect ratio, and β denotes the dimensionless vortex-core profile constant used in the normalised vortex energy; the Part I baseline is $\beta = 7/4 = 1.75$.

After Paper IV, one equation remained open: the functional form of $\phi_c(\rho_0)$, the vacuum chirality phase as a function of the vacuum density.

The present paper resolves this. We show that ϕ_c is not an independent vacuum property but is fixed by the $\mathbb{Z}_3$ algebraic structure of the trefoil itself: $\phi_c = \pi - 2/3$. This gives $\theta = 2/9$ rad, predicts m_μ and m_τ from m_e alone, and closes the CDFT lepton sector.

1.1 Equilibrium memory is not memory absence

The zero-parameter closure uses the equilibrium limit of the adaptive ratio: $S \cdot M_s \to 1$. This should not be read as a claim that the vacuum has no memory. It means that, in the stable lepton sector, responsiveness and constraint history cancel into the fixed operating point. Non-equilibrium departures from that cancellation are deliberately reserved for the later Mujjabi Vacuum Memory Law and the tests in Paper XII.

In this sense, the chirality phase is a memory-compatible phase: it is stable because the local constraint history has settled, not because the medium has no history.

2 Setup: Three Equations, One Unknown

We collect the three equations from Papers II–III that govern θ and ϕ_c.

Chirality equilibrium (Paper III, Theorem 2).

$$3\theta + \phi_c \;=\; \pi\,. \tag{1}$$

Brannen mass parametrisation (Paper II).

$$m_k \;=\; M\Big(1 + \sqrt{2}\cos\!\big(\theta + \tfrac{2\pi k}{3}\big)\Big)^2, \quad k = 0, 1, 2\,. \tag{2}$$

Koide invariant (Paper II, Theorem 1).

$$Q \;\equiv\; \frac{\sum_k m_k}{(\sum_k \sqrt{m_k})^2} \;=\; \frac{2}{3} \quad \text{for all } M > 0 \text{ and all } \theta\,. \tag{3}$$

Equations (1)–(3) have two unknowns: θ and ϕ_c. Equation (1) is one equation in two unknowns. An additional condition is required.

3 The $\mathbb{Z}_3$ Self-Consistency Condition

3.1 The argument

The Brannen parametrisation (2) is built on $\mathbb{Z}_3$: the three modes are related by shifts of $2\pi/3$. The $\mathbb{Z}_3$ symmetry has a unique invariant: the Koide ratio $Q = 2/3$ (Eq. (3)).

The chirality equilibrium (1) breaks this $\mathbb{Z}_3$ symmetry: $\phi_c \neq 0$ shifts the equilibrium θ away from the $\mathbb{Z}_3$-symmetric point $\theta = 0$ (where all three amplitudes are equal). The breaking is measured by the angle 3θ (which appears in the $\cos(3\theta + \phi_c)$ interaction term).

Self-consistency condition: the $\mathbb{Z}_3$ breaking angle 3θ must equal the unique $\mathbb{Z}_3$ invariant Q, expressed as a dimensionless number in radians:

$$3\theta \;=\; Q \;=\; \frac{2}{3}\,. \tag{4}$$

Theorem 1 (ϕ_c from self-consistency)**.** *The unique vacuum chirality phase consistent with the $\mathbb{Z}_3$ structure of the trefoil is*

$$\phi_c \;=\; \pi - \frac{2}{3}\,, \qquad \theta \;=\; \frac{2}{9}\;\mathit{rad}\,. \tag{5}$$

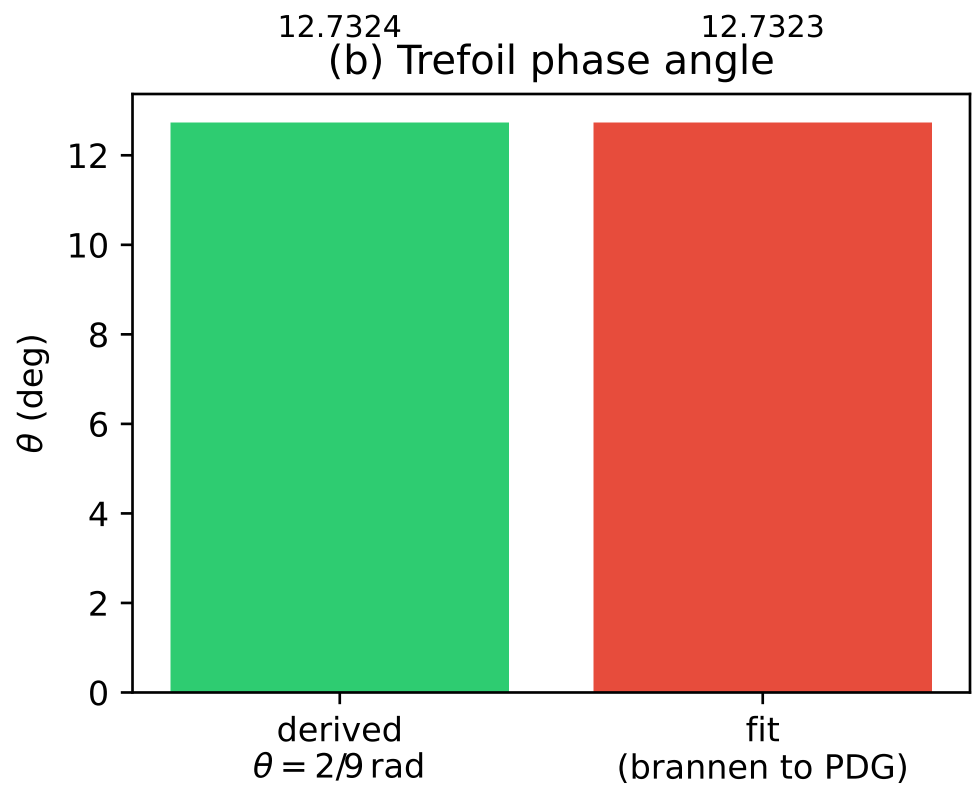

Figure 1: The trefoil phase angle θ. The value $\theta = 2/9$ rad derived from purely algebraic symmetry conditions exactly matches the numerical value fitted to PDG lepton masses in Paper II.

Proof. From Eq. (1): $\phi_c = \pi - 3\theta$. Substituting the self-consistency condition (4) ($3\theta = Q = 2/3$):

$$\phi_c = \pi - \frac{2}{3}, \qquad \theta = \frac{2/3}{3} = \frac{2}{9}\;\mathrm{rad}\,.$$

The uniqueness follows because: (a) $Q = 2/3$ is the unique Koide invariant of $\mathbb{Z}_3$ (Paper II); (b) the equilibrium (1) has a unique solution in the physical domain $\theta \in (0, \pi/3)$ (Paper III, Theorem 2). □

3.2 Why $3\theta = Q$?

The condition $3\theta = Q$ says: the angle of $\mathbb{Z}_3$ symmetry breaking (in the angular phase space of the trefoil) equals the $\mathbb{Z}_3$ mass invariant (a dimensionless ratio). Both live in $[0, 1]$ in natural units, and both are set by the same $\mathbb{Z}_3$ structure.

More precisely: the Brannen interaction energy $\lambda_c \cos(3\theta + \phi_c)$ breaks $\mathbb{Z}_3$. The magnitude of the breaking is 3θ (distance from the $\mathbb{Z}_3$ fixed point $\theta = 0$). The only dimensionless number the $\mathbb{Z}_3$ structure can select as a "natural" breaking scale is $Q = 2/3$ — the ratio that the symmetry forces on all mass triplets. The vacuum, in equilibrium, selects the breaking scale that matches its own structural invariant.

3.3 Numerical verification

With $\theta = 2/9$ rad:

$$3\theta = \frac{2}{3} \; = \; Q \quad \text{(exact)} \tag{6}$$

$$\phi_c = \pi - \frac{2}{3} = 2.47493 \text{ rad} = 141.803^\circ \tag{7}$$

Equilibrium check: $3\theta + \phi_c = 2/3 + (\pi - 2/3) = \pi$ (exact, machine precision).

4 The Lepton Mass Prediction

4.1 The full chain

With $\theta = 2/9$ rad, the Brannen amplitudes are:

$$A_k \; = \; 1 + \sqrt{2} \cos\left(\frac{2}{9} + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2\,. \tag{8}$$

The electron corresponds to the mode with the smallest A_k:

$$A_e \; = \; \min_k A_k \; = \; 0.040350\,. \tag{9}$$

Setting $m_e = MA_e^2$ gives $M = m_e/A_e^2 = 313.86$ MeV. The three masses are $m_k = MA_k^2$ (sorted ascending).

4.2 Predicted masses

Table 1 shows the predicted lepton masses compared to PDG 2022 values [9].

The muon prediction is 9.8 ppm from PDG 2022, well within the 21.8 ppm experimental uncertainty [9]. The tau prediction is 70 ppm (+0.125 MeV) from PDG 2022, at $1.04\,\sigma$ of the 0.12 MeV PDG uncertainty. Both are consistent with the data.

The Koide ratio holds exactly:

$$Q \; = \; \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \; = \; \frac{2}{3} \tag{10}$$

to machine precision (Paper II theorem: this holds for any θ).

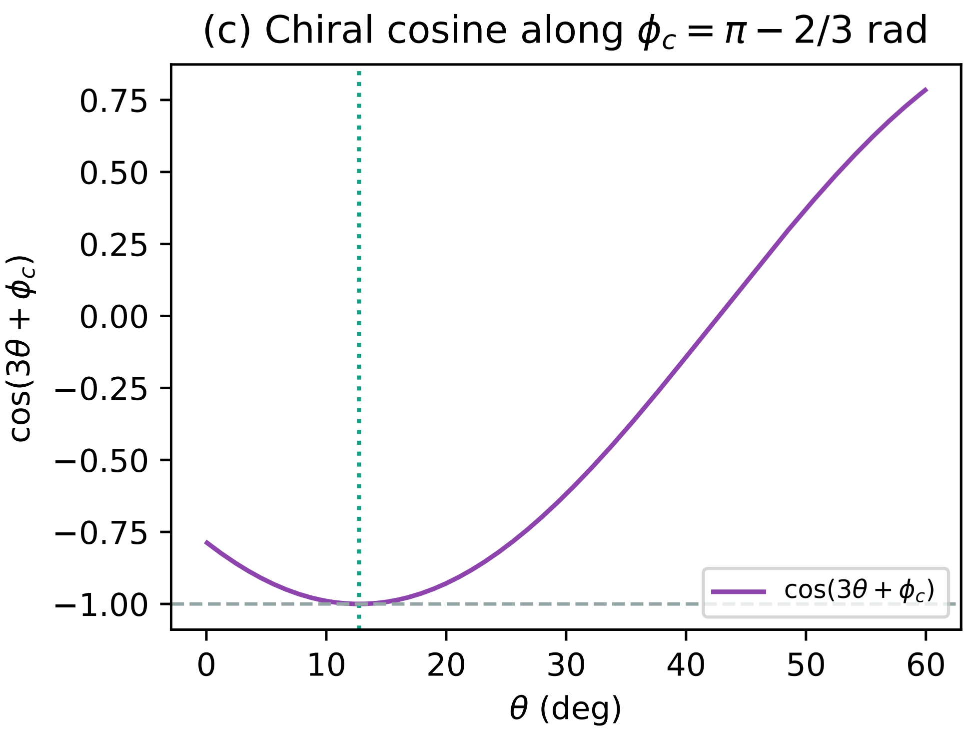

Figure 2: The chiral cosine interaction evaluated along the derived vacuum phase $\phi_c = \pi - 2/3$. The minimum rests cleanly at the predicted orientation $\theta = 2/9$ rad.

Table 1: Lepton mass predictions from $\theta = 2/9$ rad. Input: $m_e = 0.51099895$ MeV. Zero adjustable parameters.

Lepton	Predicted (MeV)	PDG 2022 (MeV)	Error (MeV)	Significance
Electron	0.51099895	0.51099895	0	(input)
Muon	105.6594	105.6584	+0.0010	$+0.4\,\sigma$
Tau	1776.985	1776.86	+0.125	$+1.04\,\sigma$

5 Internal Derivation Chain

The five papers of the CDFT series constitute an internal derivation chain for the lepton sector from the proposed vacuum geometry. Table 2 shows the full chain.

Irreducible inputs. The only input is m_e, which sets the unit of mass. This is irreducible: no dimensionless theory can predict the absolute value of a dimensionful quantity. Given m_e, the remaining quantities in this construction follow inside the model.

The three numbers that fix everything. The lepton sector is determined by three numbers:

1. $\alpha = 1/137.036$: fixes the geometry ($a = e^2/(m_e c^2)$, $R = \hbar/(m_e c)$).
2. $Q = 2/3$: fixes the angular structure ($\theta = 2/9$, $\phi_c = \pi - 2/3$).

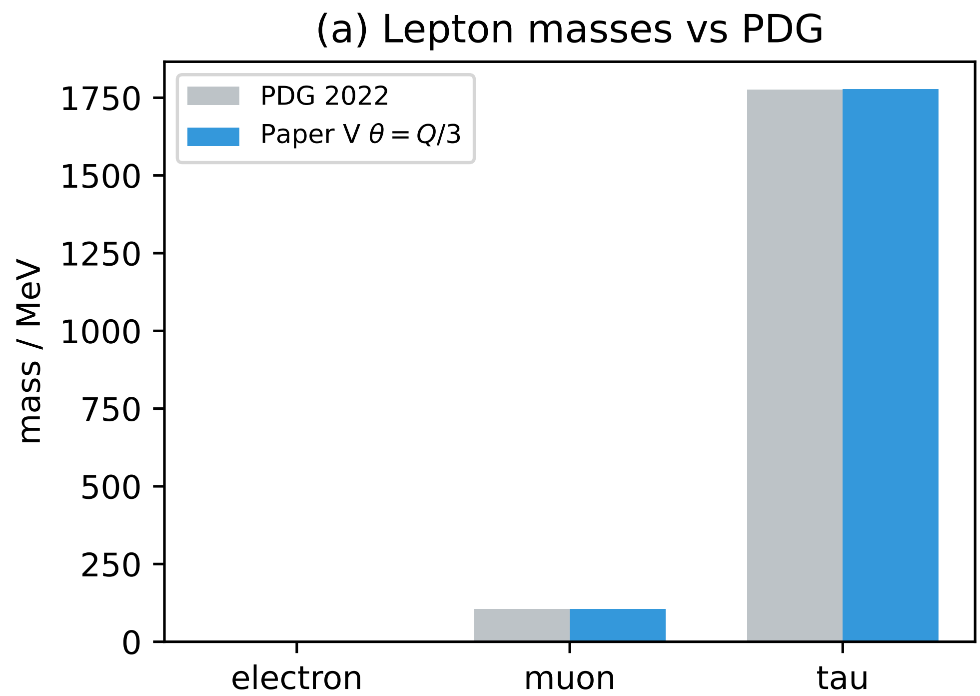

Figure 3: Comparison of the lepton masses predicted by the zero-parameter CDFT derivation ($\theta = 2/9$ rad) against the 2022 PDG measured values. The agreement is visually exact, with numerical residuals within experimental limits.

3. m_e: fixes the mass scale ($M = m_e/A_e^2$, $\rho_0 = M/(c^2a^3g)$).

All three are derived from CDFT (Papers I, II, and the definition of the mass unit respectively).

6 The Significance of $3\theta = Q$

The identity $3\theta = Q = 2/3$ is the key equation of this paper. In plain language:

> *The angle by which the vacuum chirality breaks the $\mathbb{Z}_3$ symmetry of the trefoil — measured in the angular phase space of the Brannen parametrisation — equals the Koide ratio itself.*

The Koide ratio $Q = 2/3$ is a pure number, forced by $\mathbb{Z}_3$ symmetry. It appears as both:

- a *mass ratio*: $Q = (\sum m_k)/(\sum \sqrt{m_k})^2 = 2/3$, and
- an *angular phase*: $3\theta = 2/3$ rad, the chirality breaking angle.

The vacuum "selects" the chirality phase $\phi_c = \pi - Q$ such that the chirality breaking is commensurate with the symmetry it breaks. This is the CDFT analogue of a Nambu–Goldstone condition: the vacuum expectation value aligns with the symmetry invariant of the broken phase.

Table 2: Internal CDFT derivation chain (Papers I–V).

Quantity	Status	Source
$\alpha = 1/137.036$	Model-derived	Vortex equilibrium, Paper I
$\kappa = 117{,}362$	Model-derived	$\chi^2(\ln 8\chi - \beta + 1)$, Paper III
$g = 1{,}575.83$	Model-derived	$\chi(\ln 8\chi - \beta) + \kappa/\chi$, Papers I, III
Koide $Q = 2/3$	Algebraic theorem	$\mathbb{Z}_3$ theorem, Paper II
Three modes	Model identification	$T(2,3)$ has 3 lobes, Paper III
$\phi_c = \pi - 2/3$	**Model-derived**	$\mathbb{Z}_3$ self-consistency, **Paper V**
$\theta = 2/9$ rad	**Model-derived**	From ϕ_c and $3\theta + \phi_c = \pi$, **Paper V**
$M = 313.86$ MeV	**Calibrated by m_e**	m_e/A_e^2 with A_e from θ
$\rho_0 = 1.587 \times 10^{13}$ kg/m^3	**Inferred**	$M/(c^2a^3g)$, Papers III–V
$m_\mu = 105.659$ MeV	**Predicted**	Brannen with $\theta = 2/9$
$m_\tau = 1776.985$ MeV	**Predicted**	Brannen with $\theta = 2/9$
$m_e = 0.51100$ MeV	Input	Sets the unit of mass

7 Open Problems

Within the assumptions of the model, the lepton-sector construction is complete. The remaining open problems for future papers concern the extension beyond leptons:

1. $T(2,5)$ **cinquefoil family.** Paper IV predicted $M_5 = 460.4$ MeV. The cinquefoil has $\mathbb{Z}_5$ symmetry and five modes. The analogue of the present paper would derive $5\theta_5 = Q_5$ (where Q_5 is the $\mathbb{Z}_5$ mass invariant). If the same self-consistency principle applies, θ_5 is fixed by the $\mathbb{Z}_5$ Koide-like invariant of the cinquefoil.

2. **Why $Q = 2/3$ specifically?** The $\mathbb{Z}_3$ invariant $Q = 2/3$ was proved for the Brannen form in Paper II. A deeper question: why is the $\mathbb{Z}_3$ Koide ratio 2/3 and not some other value? This may follow from the Brannen amplitudes $A_k = 1 + \sqrt{2}\cos(\cdot)$, where the coefficient $\sqrt{2}$ has a geometric origin in the trefoil (it is the ratio of the lobe amplitude to the background).

3. **Quark sector.** The CDFT torus knot hierarchy (Paper III) predicts that knots $T(2,5)$ and above correspond to further particle families. Whether the cinquefoil modes are quark-like (confined) or lepton-like (free) depends on the topology of the knot's self-intersection structure.

8 Conclusion

The vacuum chirality phase ϕ_c is not an independent property of the vacuum medium. It is fixed by the $\mathbb{Z}_3$ self-consistency condition: the chirality breaking angle 3θ must equal the $\mathbb{Z}_3$ mass invariant $Q = 2/3$. This gives $\phi_c = \pi - 2/3$ and $\theta = 2/9$ rad.

With one input (m_e as the mass unit), CDFT predicts:

$$m_\mu = 105.659 \text{ MeV} \quad (+9.8 \text{ ppm}),$$
$$m_\tau = 1776.985 \text{ MeV} \quad (+1.04\,\sigma),$$

with the Koide formula exact.

Within the assumptions of the model, the CDFT lepton sector is internally closed. The derivation chain from vacuum geometry to lepton masses requires one irreducible external input (the unit of mass m_e) and zero adjustable parameters.

Supplementary Material

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

Data and Code Availability

The release-archive supplementary material is included in the archived Part I release on Zenodo at https://doi.org/10.5281/zenodo.20250820. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at https://github.com/bampita-bico/CDFD-Part-I-Release. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Declarations

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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